

Appendix.

Table A.1 summarises the eight thematic clusters into which the 38 LOTO-RO entities are organised, together with their definitions and member classes.

Cluster	Definition	Member entities
(i) Actor	Human roles and organisational units involved in the incident	PersonPerformer, PersonParticipant, PersonSupervisor, OrganizationalUnit
(ii) Event	Dynamic sequence leading to an incident	WorkActivity, IncidentEvent, EnergyReleaseEvent
(iii) Physical	Operational environment and equipment	Plant, Machine, MachineComponent, MachineStatus, WorkAreaLocation
(iv) Risk	Hazard–risk–consequence chain	EnergySource, Hazard, Risk, Consequence
(v) Causal	Multi-level causal structure (Cause and its subtypes)	Cause, HumanError, TechnologicalFault, OrganizationalIssue, RoutineBypass, InadequateTraining, InadequateSupervision, ProcedureDeficiency, SafetyCultureIssue, ResourceOrTimePressure
(vi) Mitigation	Preventive and reactive safety measures	ControlMeasure, ProtectiveDevice, ProtectiveProcedure, AidOrTreatment
(vii) Procedural	Formal structure of LOTO procedures and isolation devices	LOTOProcedure, LOTOProcedureStep, LOTOREquirement, LOTODevice, Authorization
(viii) Requirement	Regulatory and compliance dimensions	Violation, Penalty, ActivityRequirement

Table A.1. Ontology entity clusters and their member entities

Table A.2 reports illustrative Cypher queries executed against the sample Neo4j KG, comparing system responses with gold-standard answers used for CQ validation at the data level. For CQs S3 and S4, the Cypher query returns an empty result set, as no matching instances exist in the sample KG; in these cases, the system's "I don't know the answer" response is treated as equivalent to the gold response "No one", since both correctly indicate the absence of matching entities rather than a retrieval failure.

CQ ID	Retrieved Context (Cypher Output)	Generated Response	Gold Answer	Standard
S1	[{'p.Role': 'employee', 'violations': 2}]	Employee roles most frequently violate LOTO procedures, with 2 violations.	Employee	with 2 violations
S2	[{'p.name': 'Manufacturing', 'fatalities': 3}]	The Manufacturing plant has the highest number of fatalities, with 3 fatalities.	Manufacturing	with 3 fatal accidents
S3	[]	I don't know the answer.	No one	
S4	[]	I don't know the answer.	No one	
B1	[]	No.	No	
B2	[]	No.	No	
B3	[{'ie.IncidentTime': '15:09', 'c.name': 'death'}, {'ie.IncidentTime': '22:00', 'c.name': 'death'}, {'ie.IncidentTime': '12:00 a.m.', 'c.name': 'death'}, {'ie.IncidentTime': '14:45', 'c.name': 'fatal injuries'}]	Yes.	Yes	
B4	[{'mechanicalAccidents': 3}]	Yes.	Yes (3 su 4)	
C1	[{'count(i)': 0}]	0	0	
C2	[{'count(i)': 1}]	1	1 in Services	
C3	[{'count(ie)': 0}]	0	0	
C4	[{'count(ie)': 1}]	1	1	

Table A.2. Results of Cypher queries executed on a sample KG populated in Neo4j